

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MEC-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **24** pages.

E6

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.
Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
8. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
9. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
10. No candidate, without special permission of the Invigilator, would leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
14. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
16. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ **Invigilator's Signature :** _____

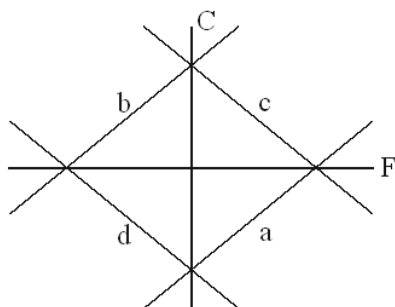
Your Target is to secure Good Rank in Pre-Medical 2024

Topic : MAGNETIC FIELD, THERMAL PHYSICS, THERMOMETRY & THERMAL EXPANSION, MODERN PHYSICS (RADIOACTIVITY), SEMICONDUCTOR [BACK UNIT] : WAVE OPTICS AND UNIT DIMENSION ERROR

TG: @Chalnaayaaar

SECTION-A (PHYSICS)

1. Which of the curves in figure represents the relation between Celsius and Fahrenheit temperature?



- (1) Curve a (2) Curve b
(3) Curve c (4) Curve d
2. Relation between α , β & γ for isotropic body is :-
(1) $\alpha : \beta : \gamma = 3 : 2 : 1$
(2) $\alpha : \beta : \gamma = 2 : 1 : 3$
(3) $\alpha : \beta : \gamma = 1 : 2 : 3$
(4) $\alpha : \beta : \gamma = 3 : 1 : 2$
3. If brass and steel rod having thermal coefficient α_1 and α_2 and length of brass and steel ℓ_1 and ℓ_2 if change in length ($\ell_2 - \ell_1$) is same for any temperature change than which is correct :
(1) $\alpha_1 \ell_2 = \alpha_2 \ell_1$ (2) $\alpha_1 \ell_2^2 = \alpha_2 \ell_1^2$
(3) $\alpha_1 \ell_2 = \alpha_2^2 \ell_2^2$ (4) $\alpha_1 \ell_1 = \alpha_2 \ell_2$
4. An iron rod connect small heavy bob of pendulum clock. Find fraction loss in time period of pendulum clock if temperature is increased from 10°C to 30°C ($\alpha_{\text{Fe}} = 5 \times 10^{-6} \text{C}^{-1}$)
(1) 5×10^{-5} (2) 2×10^{-5}
(3) 1×10^{-5} (4) 8×10^{-5}
5. In YDSE, the intensity of light at a point on the screen where the path difference is λ is I_0 . The intensity at a point where the path difference is $\lambda/3$, will be :-
(1) $\frac{I_0}{3}$ (2) $\frac{I_0}{4}$ (3) I_0 (4) $\frac{I_0}{2}$

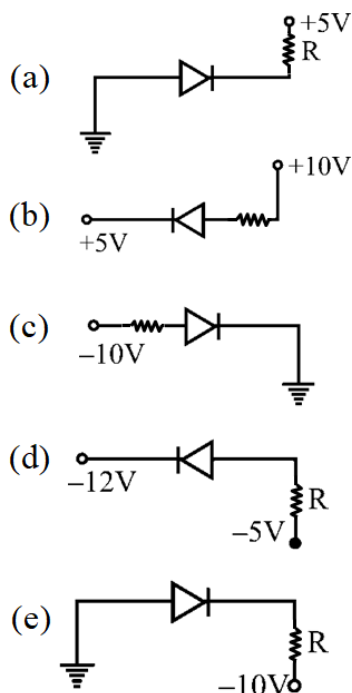
6. If the 8th bright band due to light of wavelength λ_1 coincides with 9th bright band from light of wavelength λ_2 in Young's double slit experiment, then the possible wavelengths of visible light are :-
(1) 400 nm and 450 nm
(2) 425 nm and 400 nm
(3) 400 nm and 425 nm
(4) 450 nm and 400 nm
7. A light source of $5000\text{\AA}$ wave length produces a single slit diffraction. The first minima in diffraction pattern is seen, at a distance of 5mm from central maxima. The distance between screen and slit is 2 metre. The width of slit in mm will be :
(1) 0.1 (2) 0.4 (3) 0.2 (4) 2
8. Wavelengths of light used in an optical instrument are $\lambda_1 = 4000 \text{\AA}$ and $\lambda_2 = 5000\text{\AA}$, then ratio of their respective resolving power (corresponding to λ_1 and λ_2) is:-
(1) 16 : 25 (2) 9 : 1
(3) 4 : 5 (4) 5 : 4
9. When light ray moving from air to medium is incident at polarising angle then
(a) Reflected light is completely polarised
(b) Refracted light is partial polarised
(c) Both reflected and refracted light are perpendicular to each other.
(d) Brewster's Law $\mu = \tan i_p$ is obeyed
Choose the most correct option.
(1) a and d are correct
(2) a, c, d are correct
(3) a, b, c, d are correct
(4) d is correct

TG: @Chalnaayaaar

PHASE - MEC

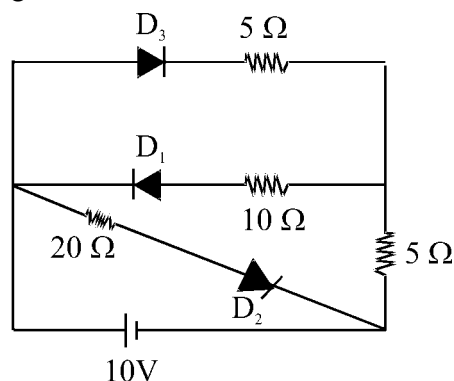
10. Consider the following reaction
 ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^4_2\text{He} + \text{Q}$
 If $m({}^2_1\text{H}) = 2.0141\text{u}$; $m({}^4_2\text{He}) = 4.0024\text{u}$, the energy Q released (in MeV) in this fusion reaction is :-
 (1) 12 (2) 6 (3) 24 (4) 48
11. The charges on beta particles(β) alpha particles (α) and gamma (γ) rays are
 (1) +2, +1, 0 (2) +1, -1, +2
 (3) -1, +2, 0 (4) -1, -2, 0
12. Consider the nuclear reactions
 ${}_2\text{He}^4 + {}_Z\text{X}^A \rightarrow {}_{Z+2}\text{Y}^{A+3} + \text{W}$.
 Then, W denotes :
 (1) electron
 (2) proton
 (3) positron
 (4) neutron
13. When ${}_{92}\text{U}^{238}$ changes into ${}_{82}\text{Pb}^{206}$, then the number of α and β -particles emitted are :
 (1) 6 and 6 (2) 8 and 8
 (3) 6 and 8 (4) 8 and 6
14. A radioactive element X decays into another stable element Y. Half life of X is 2hrs. Initially only X is present. After time t the ratio of atoms of X and Y is found to be 1 : 3, then t in hours is :-
 (1) 2 (2) 4
 (3) 5 (4) 6
15. Plutonium has atomic mass 210 and a decay constant equal to $5.8 \times 10^{-8} \text{ s}^{-1}$. The number of α -particle emitted per second by 1 mg is - (Avogadro's Constant = 6.0×10^{23})
 (1) 1.7×10^9
 (2) 1.7×10^{11}
 (3) 2.9×10^{11}
 (4) 3.4×10^9

16. Which statement is correct :-
 (1) N-type germanium is negatively charged and P-type germanium is positively charged
 (2) Both N-type and P-type germanium are neutral
 (3) N-type germanium is positively charged and P-type germanium is negatively charged
 (4) Both N-type and P-type germanium are negatively charged
17. In the given figure, which of the diodes are forward biased ?



- (1) a, b, c (2) b, d, e
 (3) a, c, d (4) b, c, d

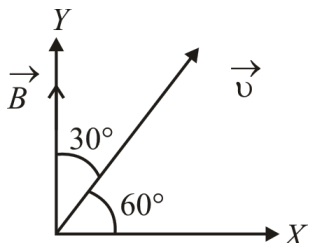
18. In the given circuit



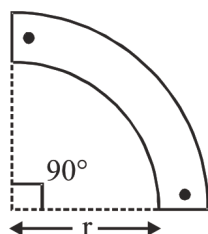
The current through the battery is

- (1) 0.5A (2) 1 A (3) 1.5 A (4) 2 A

19. A proton of mass $1.67 \times 10^{-27} \text{ kg}$ and charge $1.6 \times 10^{-19} \text{ C}$ is projected with a speed of $2 \times 10^6 \text{ m/s}$ at an angle of 60° to the X-axis. If a uniform magnetic field of 0.104 Tesla is applied along Y-axis, the path of proton is



- (1) A circle of radius 0.2 m and time period $\pi \times 10^{-7} \text{ s}$
 (2) A circle of radius 0.1 m and time period $2\pi \times 10^{-7} \text{ s}$
 (3) A helix of radius 0.1 m and time period $2\pi \times 10^{-7} \text{ s}$
 (4) A helix of radius 0.2 m and time period $4\pi \times 10^{-7} \text{ s}$
20. A bar magnet of length ' ℓ ' and magnetic dipole moment ' M ' is bent in the form of an arc as shown in figure. The new magnetic dipole moment will be



- (1) $\frac{\sqrt{2}M}{\pi}$ (2) M
 (3) $\frac{3}{\pi} M$ (4) $2\sqrt{2} \frac{M}{\pi}$
21. The magnetic field strength at a point at a distance ' d ' from the centre, on the axial line of a very short bar magnet of magnetic moment M is B . The magnetic induction at a distance $2d$ from the centre, on the equatorial line of a magnet of magnetic moment $8M$, will be :

- (1) $4B$ (2) $\frac{B}{2}$ (3) $\frac{B}{4}$ (4) $2B$

22. A magnetic needle suspended parallel to a magnetic field requires $\sqrt{3} \text{ J}$ of work to turn it through 60° . The torque needed to maintain the needle in this position will be :-

- (1) $\sqrt{3} \text{ N-m}$ (2) $\frac{3}{2} \text{ N-m}$
 (3) $2\sqrt{3} \text{ N-m}$ (4) 3 N-m

23. An electron having mass ($9.1 \times 10^{-31} \text{ kg}$) and charge ($1.6 \times 10^{-19} \text{ C}$) moves in a circular path of radius 0.5 m with a velocity 10^6 m/s in a magnetic field. Strength of magnetic field is :-

- (1) $1.13 \times 10^{-5} \text{ T}$ (2) $5.6 \times 10^{-6} \text{ T}$
 (3) $2.8 \times 10^{-6} \text{ T}$ (4) None of these

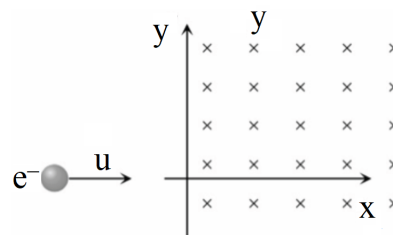
24. A nonconducting disc of radius R is rotating about an axis passing through its centre and perpendicular to its plane with an angular velocity ω . Charge q is uniformly distributed over its surface. The magnetic moment of the disc is

- (1) $\frac{1}{4} q \omega R^2$ (2) $\frac{1}{4} q \omega R$
 (3) $q \omega R$ (4) $\frac{1}{2} q \omega R^2$

25. An electron having kinetic energy K is moving in a circular orbit of radius R perpendicular to a uniform magnetic induction. If kinetic energy is doubled and magnetic induction tripled, the radius will become

- (1) $\frac{2R}{3}$ (2) $\frac{\sqrt{2}}{3} R$ (3) $\sqrt{\frac{2}{3}} R$ (4) $\frac{2}{\sqrt{3}} R$

26. An electron moving with a speed u along the positive x-axis at $y = 0$ enters a region of uniform magnetic field $\vec{B} = -B_0 \hat{k}$ which exists to the right of y-axis. The electron exits from the region after some time with the speed v at co-ordinate y , then

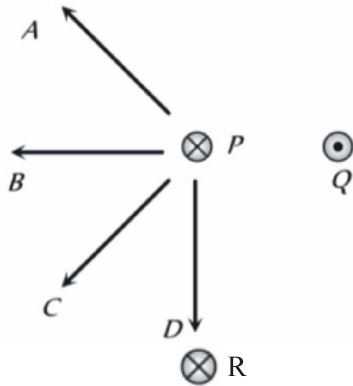


- (1) $v > u, y < 0$ (2) $v = u, y > 0$
 (3) $v > u, y > 0$ (4) $v = u, y < 0$

27. A particle of mass M and charge Q moving with velocity $\vec{v}$ describes a circular path of radius R when subjected to a uniform transverse magnetic field of induction B . The work done by the field when the particle completes one full circle is

- (1) $BQv2\pi R$ (2) $\left(\frac{Mv^2}{R}\right)2\pi R$
 (3) Zero (4) $BQ2\pi R$

28. In figure shows three long straight wires P, Q and R carrying currents normal to the plane of the paper. All three currents have the same magnitude. Which arrow best shows the direction of the resultant force on the wire P



- (1) A (2) B (3) C (4) D

29. A charge of $2.0 \mu\text{C}$ moves with a speed of $3.0 \times 10^6 \text{ ms}^{-1}$ along +ve X-axis. A magnetic field of strength $\vec{B} = -0.2\hat{k}\text{ Tesla}$ exists in space. What is the magnetic force ($\vec{F}_m$) on the charge

- (1) $F_m = 1.2 \text{ N}$ along +ve x-direction
 (2) $F_m = 1.2 \text{ N}$ along -ve x-direction
 (3) $F_m = 1.2 \text{ N}$ along +ve y-direction
 (4) $F_m = 1.2 \text{ N}$ along -ve y-direction

30. If the error in the measurement of radius of circle is 3% then calculate the error in measurement of area of circle :-

- (1) 5 % (2) 8 %
 (3) 3 % (4) 6 %

31. If $x = at + bt^2$, where x is the distance travelled by the body in kilometres while t is the time in seconds, then the unit of b is :-

- (1) km/s (2) km-s
 (3) km/s^2 (4) km-s^2

32. If force (f), velocity (v) and time(t) are fundamental quantities then find out dimensional formula of torque (τ)

- (1) $F^2V^1T^1$
 (2) $F^1V^1T^1$
 (3) FV^2T^{-2}
 (4) $FV^{-1}T^{-1}$

33. The main scale of vernier caliper is having x divisions per cm, x divisions of the vernier scale coincide with $(x - 2)$ division of main scale. The least count of vernier caliper is

- (1) $\frac{2}{(x+1)(x-1)}\text{cm}$
 (2) $\frac{2}{x}\text{cm}$
 (3) $\frac{2}{x^2}\text{cm}$
 (4) $\frac{2}{x(x+1)}\text{cm}$

34. A screw gauge has least count 0.01 mm & there are 100 divisions in circular scale. The pitch of screw gauge is

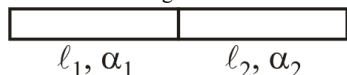
- (1) 1 mm
 (2) 2 mm
 (3) 0.1 mm
 (4) 0.2 mm

35. The dimensions of mutual inductance is

- (1) $M^1L^2T^{-2}A^{-2}$
 (2) $M^1LT^{-3}A^{-1}$
 (3) $M^1L^2T^{-1}A^{-2}$
 (4) $M^1L^3T^{-2}A^{-1}$

SECTION-B (PHYSICS)

36. Two rods of length ℓ_1 and ℓ_2 having coefficient of thermal expansion as α_1 and α_2 respectively are placed such that their one end is common as shown in figure. The temperature of system is raised by ΔT . Find α_{avg} of composite system.



- (1) $\frac{\ell_1 \alpha_1 + \ell_2 \alpha_2}{\ell_1 + \ell_2}$ (2) $\frac{\ell_1 \alpha_1 - \ell_2 \alpha_2}{\ell_1 + \ell_2}$
 (3) $\frac{\ell_2 \alpha_2 - \ell_1 \alpha_1}{\ell_1 + \ell_2}$ (4) None of these
37. Find the thickness of a plate which will produce a change in optical path equal to half the wavelength λ of the light passing through it, normally. The refractive index of the plate is μ :-
- (1) $\frac{\lambda}{(\mu - 1)}$ (2) $\frac{(\mu - 1)\lambda}{2}$
 (3) $\frac{\lambda}{2(\mu - 1)}$ (4) $(\mu - 1)\lambda$
38. Unpolarised light of intensity 32 Wm^{-2} passes through three polarizers such that transmission axes of the first and second polarizer makes an angle 30° with each other and the transmission axis of the last polarizer is crossed with that of the first. The intensity of final emerging light will be
- (1) 32 Wm^{-2} (2) 3 Wm^{-2}
 (3) 8 Wm^{-2} (4) 4 Wm^{-2}
39. A gamma ray photon creates an electron-positron pair. If the rest mass energy of an electron is 0.5 MeV and the total K.E. of the electron- positron pair is 0.78 MeV , then the energy of the gamma ray photon must be :-
- (1) 0.78 MeV (2) 1.78 MeV
 (3) 1.28 MeV (4) 0.28 MeV

40. Fusion reaction takes place at high temperature because :-

- (1) Molecules break up at high temperature
 (2) Nuclei break up at high temperature
 (3) Atoms get ionised at high temperature
 (4) Kinetic energy is high enough to overcome the coulomb repulsion between nuclei

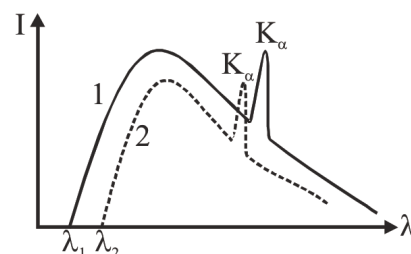
41. The half life of a radioactive substance against a decay is $1.2 \times 10^7 \text{ s}$. What is the decay rate for 4.0×10^{15} atoms of the substance ?

- (1) $4.6 \times 10^{12} \text{ atoms/s}$
 (2) $2.3 \times 10^{11} \text{ atoms/s}$
 (3) $4.6 \times 10^{10} \text{ atoms/s}$
 (4) $2.3 \times 10^8 \text{ atoms/s}$

42. After 30 days, the activity of a radioactive sample is 2000 Bq . After another 15 days, the activity reduces to 1000 Bq . The initial activity of the sample will be :-

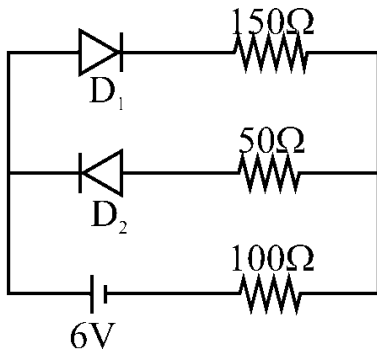
- (1) 8000 Bq (2) 4000 Bq
 (3) 2000 Bq (4) 500 Bq

43. The intensity distribution of X-rays from two Coolidge tubes operated on different voltages V_1 and V_2 and using different target materials of atomic numbers Z_1 and Z_2 is shown in the figure. Which one of the following inequalities is true ?

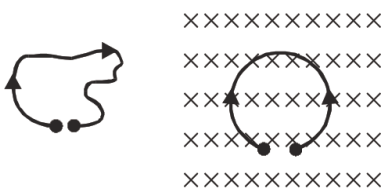


- (1) $V_1 > V_2, Z_1 < Z_2$
 (2) $V_1 > V_2, Z_1 > Z_2$
 (3) $V_1 < V_2, Z_1 > Z_2$
 (4) $V_1 = V_2, Z_1 < Z_2$

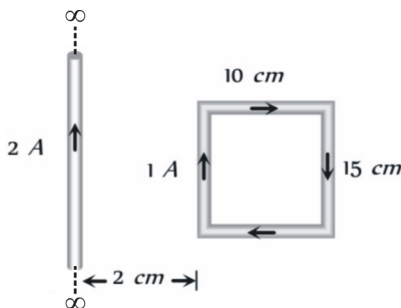
44. The circuit shown in following figure contains two diode D_1 and D_2 each with a forward resistance of $50\ \Omega$ and with infinite backward resistance. If the battery voltage is 6 V , the current through the $100\ \Omega$ resistance (in amperes) is :-



- (1) Zero (2) 0.02 (3) 0.03 (4) 0.036
45. A thin flexible wire of length L is connected to two adjacent fixed points carries a current I in the clockwise direction, as shown in the figure. When system is put in a uniform magnetic field of strength B going into the plane of paper, the wire takes the shape of a circle. The tension in the wire is:

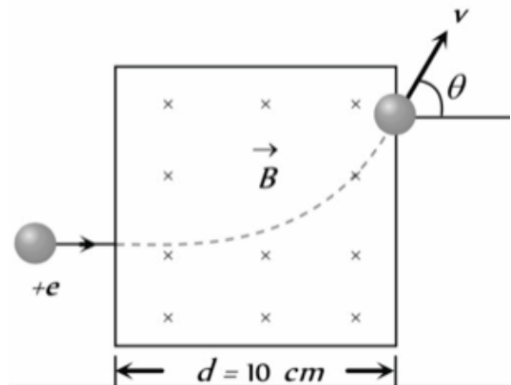


- (1) IBL (2) $\frac{IBL}{\pi}$
 (3) $\frac{IBL}{2\pi}$ (4) $\frac{IBL}{4\pi}$
46. What is the net force on the square coil

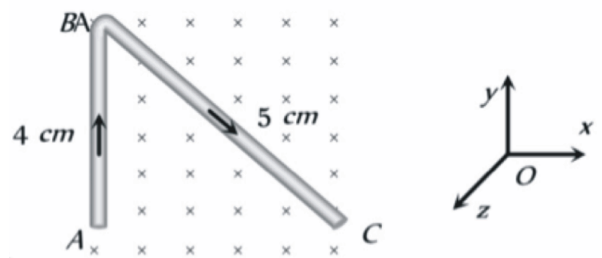


- (1) $25 \times 10^{-7}\text{ N}$ moving towards wire
 (2) $25 \times 10^{-7}\text{ N}$ moving away from wire
 (3) $35 \times 10^{-7}\text{ N}$ moving towards wire
 (4) $35 \times 10^{-7}\text{ N}$ moving away from wire

47. A proton accelerated by a potential difference 500 KV moves through a transverse magnetic field of 0.51 T as shown in figure. The angle θ through which the proton deviates from the initial direction of its motion is



- (1) 15° (2) 30° (3) 45° (4) 60°
48. A uniform conducting wire ABC has a mass of 10 g . A current of 2 A flows through it. The wire is kept in a uniform magnetic field $B = 2\text{ T}$. The acceleration of the wire will be

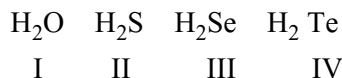


- (1) Zero
 (2) 12 ms^{-2} along y-axis
 (3) $1.2 \times 10^{-3}\text{ ms}^{-2}$ along y-axis
 (4) $0.6 \times 10^{-3}\text{ ms}^{-2}$ along y-axis
49. Dimension of resistivity (ρ) :
- (1) $[M L^3 T^{-3} A^{-1}]$ (2) $[M L^3 T^{-3} A^{-2}]$
 (3) $[M L^{-3} T^2 A^2]$ (4) $[M L T^{-3} A^{-1}]$
50. In an experiment, the % error occurred in measurement of physical quantities A, B, C & D are 1%, 2%, 4% & 3% respectively. Then the maximum percentage error in the measurement of X, where $X = \frac{A^2 B^{1/2}}{C^{1/2} D^3}$ is:
- (1) 12% (2) 16% (3) 14% (4) 7%

Topic : OXYGEN CONTAINING, NITROGEN CONTAINING COMPOUND, CHEMICAL BONDING, s-BLOCK, BIOMOLECULE, POLYMER, CHEMICAL EQUILIBRIUM TILL TAUGHT [BACK UNIT] : NOMENCLATURE, ISOMERISM

SECTION-A (CHEMISTRY)

51. Consider the following molecules ;



Arrange these molecules in increasing order of bond angles.

- (1) $\text{I} < \text{II} < \text{III} < \text{IV}$ (2) $\text{IV} < \text{III} < \text{II} < \text{I}$
 (3) $\text{I} < \text{II} < \text{IV} < \text{III}$ (4) $\text{II} < \text{IV} < \text{III} < \text{I}$

52. Molecule containing $p\pi-d\pi$ bond from the given choices is

- (1) ClO_4^- (2) H_2O (3) NO_3^- (4) CO_3^{2-}

53. Correct order of volatility is:-

- (1) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
 (2) $\text{HCl} > \text{HBr} > \text{HI} > \text{HF}$
 (3) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
 (4) $\text{HBr} < \text{HCl} < \text{HI} < \text{HF}$

54. Which of the following pairs have identical values of bond order ?

- (1) N_2^+ and O_2^+ (2) F_2 and Ne_2
 (3) O_2 and B_2 (4) C_2 and N_2

55. Correct order of polarisation (M = alkali metals)

- (1) $\text{MF} > \text{MBr} > \text{MCl} > \text{MI}$
 (2) $\text{MI} > \text{MBr} > \text{MCl} > \text{MF}$
 (3) $\text{MI} > \text{MBr} > \text{MF} > \text{MCl}$
 (4) $\text{MF} > \text{MCl} > \text{MBr} > \text{MI}$

56. The product obtained on heating LiNO_3 will be :

- (1) $\text{LiNO}_2 + \text{O}_2$ (2) $\text{Li}_2\text{O} + \text{NO}_2 + \text{O}_2$
 (3) $\text{Li}_3\text{N} + \text{O}_2$ (4) $\text{Li}_2\text{O} + \text{NO} + \text{O}_2$

57. Which metal does not form ionic hydride –

- (1) Na (2) K (3) Ca (4) Be

58. Which of the following statement are correct :

- (a) Superoxides are paramagnetic and coloured
 (b) gypsum is added in cement to decrease setting time of cement.
 (c) NaHCO_3 is use in fire extinguishers
 (d) For good quality cement ratio of silica to Alumina should be between 1.5 to 2.5
 (e) in normal human body order of metal is $\text{Ca} > \text{Na} > \text{Fe} > \text{Mg} > \text{K}$

- (1) a, b, c (2) b, c, e (3) a, c (4) a, c, e

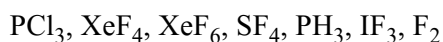
59. Correct decreasing order of dipole moment.

- (1) $\text{H}_2\text{S} > \text{H}_2\text{O} > \text{NH}_3 > \text{CH}_4$
 (2) $\text{CH}_4 > \text{H}_2\text{O} > \text{H}_2\text{S} > \text{NH}_3$
 (3) $\text{H}_2\text{O} > \text{NH}_3 > \text{H}_2\text{S} > \text{CH}_4$
 (4) $\text{NH}_3 > \text{H}_2\text{O} > \text{H}_2\text{S} > \text{CH}_4$

60. Which carbonate kept in atmosphere of CO_2 .

- (1) BeCO_3 (2) MgCO_3
 (3) CaCO_3 (4) BaCO_3

61. How many molecules are non-polar in given molecules :



- (1) 4 (2) 5 (3) 2 (4) 3

62. In which hydration energy is greater than lattice energy.

- (1) BeSO_4 (2) CaSO_4
 (3) SrSO_4 (4) BaSO_4

63. For a gaseous reaction, $3\text{H}_2(\text{g}) + \text{N}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, the partial pressures of H_2 and N_2 are 0.4 and 0.8 atmospheric at equilibrium respectively while the total pressure of the entire system is 2.8 atmospheric. What will be the value of K_p :

- (1) 32 atm^{-2} (2) 20 atm^{-2}
 (3) 50 atm^{-2} (4) 80 atm^{-2}

TG: @Chalnaayaaar

PHASE - MEC

1015CMD303361230059

14-08-2023

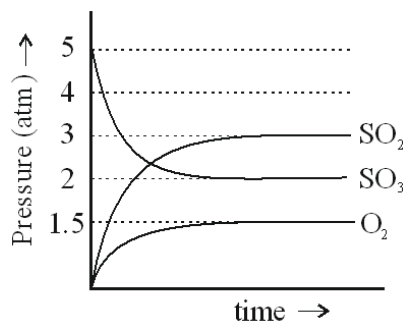
64. For the reaction $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$ if initial pressure is 4 atm and at equilibrium total pressure becomes 5 atm, value of K_P will be :

- (1) 0.33 (2) 1.33
(3) 0.67 (4) 1.67

65. $\text{NH}_4\text{HS}_{(s)} \rightleftharpoons \text{NH}_{3(g)} + \text{H}_2\text{S}_{(g)}$; Initially NH_4HS is taken in a closed container and total pressure at equilibrium is 6 atm then find K_P for above reaction :

- (1) 6 (2) 3 (3) 9 (4) 36

66. The dissociation of $\text{SO}_3(\text{g})$ into $\text{SO}_2(\text{g})$ and $\text{O}_2(\text{g})$ is carried out in a closed vessel at constant temperature T. The K_P is 'x' atm^2 . The partial pressure variation for the gases is shown in the graph. The value of 'x' is :



- (1) $\frac{3^{3/2}}{2^{3/2}}$ (2) 2^2
(3) 1 (4) $\frac{3^3}{2^3}$

67. Select the incorrect statement-

- (1) Equilibrium constant is temperature dependent having one unique value for a particular reaction represented by a balanced equation
- (2) Equilibrium constant of reverse reaction is equal to the inverse of the equilibrium constant for the forward reaction.
- (3) If $K_c > 10^3$, the reaction proceeds nearly to completion.
- (4) If $Q < K$ the reaction will proceed in backward direction

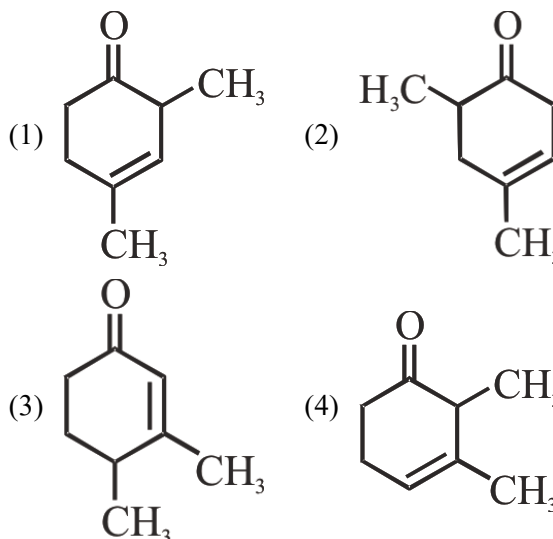
68. Match list-I with list-II

	List-I		List-II
(A)	$\text{Ni(s)} + 4\text{CO(g)} \rightleftharpoons \text{Ni(CO)}_4(\text{g})$	(I)	$K_P = K_C$
(B)	$2\text{NOCl(g)} \rightleftharpoons 2\text{NO(g)} + \text{Cl}_2(\text{g})$	(II)	$4\text{NO} + 6\text{H}_2\text{O} \rightleftharpoons 4\text{NH}_3 + 5\text{O}_2$
(C)	$K_C = \frac{[\text{NH}_3]^4[\text{O}_2]^5}{[\text{NO}]^4[\text{H}_2\text{O}]^6}$	(III)	$K_P = K_C (\text{RT})^{-3}$
(D)	$\text{CH}_4(\text{g}) + 2\text{S}_2(\text{g}) \rightleftharpoons \text{CS}_2(\text{g}) + 2\text{H}_2\text{S(g)}$	(IV)	$K_C = K_P(\text{RT})^{-1}$

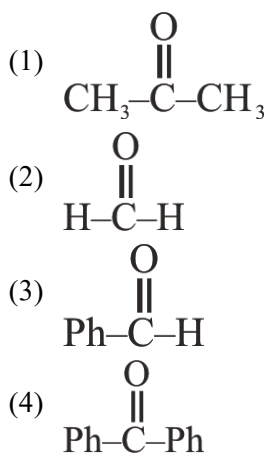
Choose the correct answer from the options given below :-

- (1) A-IV, B-II, C-I, D-III (2) A-III, B-IV, C-II, D-I
(3) A-III, B-I, C-IV, D-II (4) A-IV, B-I, C-II, D-III

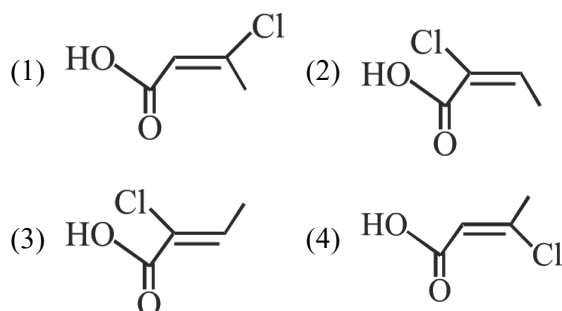
69. 2,4-Dimethyl-3-cyclohexenone can be written as :-



70. Which of the following can form two oximes on reaction with $\text{NH}_2\text{-OH}$?



71. (Z)3-Chlorobut-2-enoic acid is :

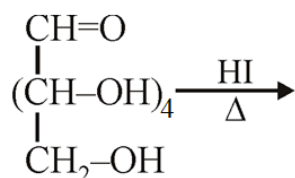


72. Which of the following is/are biodegradable polymer.

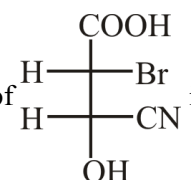
- (a) PHBV
 (b) Nylon (6, 6)
 (c) Nylon-2-Nylon-6

- (1) a, b, c (2) a, c (3) a, b (4) b, c

73. Major product of given reaction is

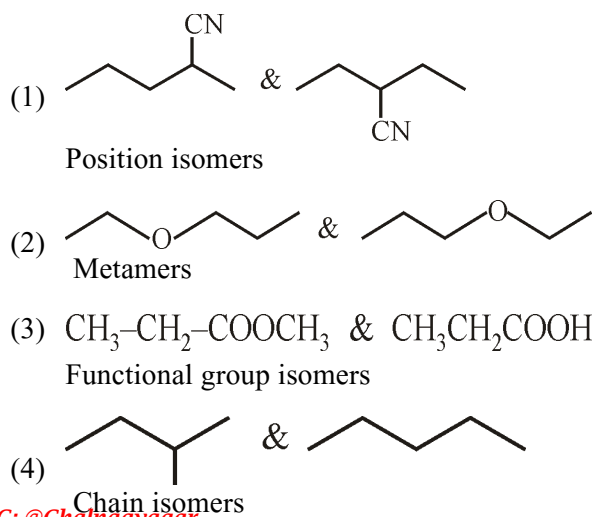


- (1) n-Pentane (2) Hexan-1-ol
 (3) n-Hexane (4) Hexanal

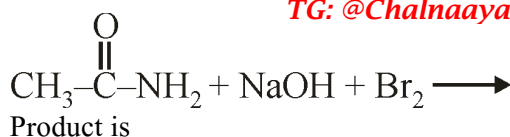
74. The absolute configuration of  is :-

- (1) 2R, 3S (2) 2R, 3R
 (3) 2S, 3S (4) 2S, 3R

75. Correct relation of isomers is :-

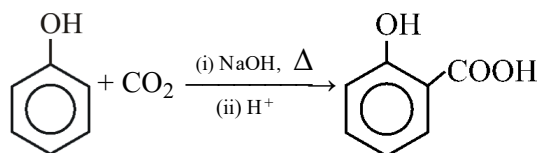


76.



- (1) $\text{CH}_3\text{-CH}_2\text{-NH}_2$ (2) $\text{CH}_3\text{-NH}_2$
 (3) $\text{CH}_3\text{-N}_2^+$ (4) $\text{CH}_3\text{-CH}_2\text{-OH}$

77.



The above reaction is

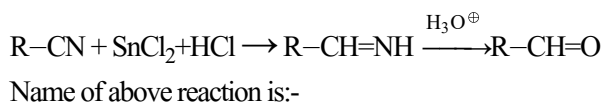
- (1) Kolbe's electrolysis
 (2) Reimer-Tiemann reaction
 (3) Etard Reaction
 (4) Kolbe Reaction

78. Major product is



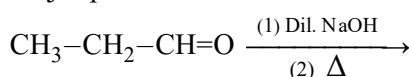
- (1) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CHO}$
 (2) $\text{CH}_3\text{-CH=CH-CH}_2\text{-CHO}$
 (3) $\text{CH}_3\text{-CH=CH-CHO}$
 (4) None

79.



- (1) Stephen reduction
 (2) Rosenmund reduction
 (3) Bouveault blanc reduction
 (4) Mendius reduction

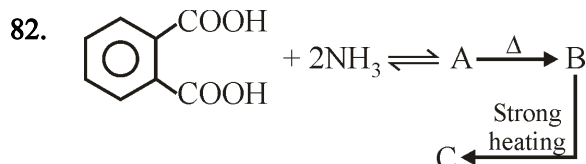
80. Major product is



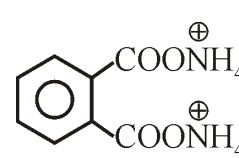
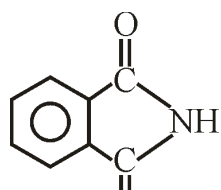
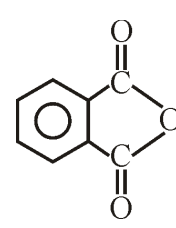
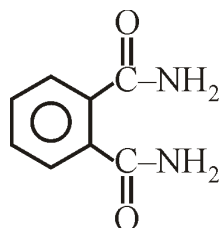
- (1) $\text{CH}_3\text{-CH}_2\text{-CH=CH-CH}_2\text{-CH=O}$
 (2) $\text{CH}_3\text{-CH}_2\text{-CH=C(CH}_3\text{)-CH=O}$
 (3) $\text{CH}_3\text{-CH=CH-CHO}$
 (4) $\text{CH}_3\text{-C(CH}_3\text{)=C(CH}_3\text{)-CHO}$

81. Correct reaction is:

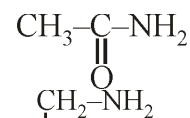
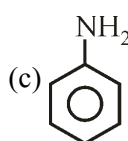
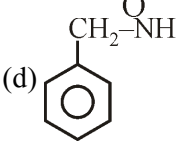
- (1) $2\text{HCHO} \xrightarrow{\ominus\text{OH}} \text{CH}_2=\text{CH}-\text{CH}=\text{O}$
 (2) $2\text{Ph}-\text{CH}=\text{O} \xrightarrow{\ominus\text{OH}} \text{Ph}-\text{CH}_2-\text{OH} + \text{PhCOO}^\ominus$
 (3) $\begin{array}{c} \text{CHO} \\ | \\ \text{CHO} \end{array} \xrightarrow{\ominus\text{OH}} \begin{array}{c} \text{COO}^\ominus \\ | \\ \text{COO}^\ominus \end{array}$
 (4) $\text{CH}_3-\text{CH}=\text{O} \xrightarrow{\ominus\text{OH}} \text{CH}_3-\text{CH}_2-\text{OH} + \text{CH}_3-\text{COO}^\ominus$



What is C ?

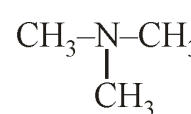
- (1)  (2) 
 (3)  (4) 

83. Which of the following give carbyl amine test positive ?

- (a) $\text{CH}_3-\text{CH}_2-\text{NH}_2$ (b) 
 (c)  (d) 

- (1) a only (2) a, d only
 (3) a, b, c, d (4) a, c, d only

84. Which of the following react with Hinsberg reagent and the product is not soluble in base.

- (1) $\text{CH}_3-\text{CH}_2-\text{NH}_2$
 (2) $\text{CH}_3-\text{NH}-\text{CH}_3$
 (3) 
 (4) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{NH}_2$

85. Which of the following is NOT true statement about amylose ?

- (1) It is water insoluble
 (2) Amylose is a long unbranched chain of α -D glucose units held together by C_1 - C_4 glycosidic linkage.
 (3) It is 15-20% of starch
 (4) All are incorrect

SECTION-B (CHEMISTRY)

86. N-N bond length is minimum in :

- (1) N_2O (2) N_2O_3 (3) N_2O_4 (4) N_2O_5

87. Number of σ and π bonds present in SO_3 as per octet rule are.

- (1) $3\sigma, 3\pi$ (2) $3\sigma, 2\pi$
 (3) $3\sigma, 1\pi$ (4) $3\sigma, 0\pi$

88. Which amongst following is incorrect statement ?

- (1) according to MOT in C_2^- only π bond is present
 (2) O_2^+ and O_2^- is paramagnetic
 (3) Stability of B_2 is more than F_2
 (4) Bond order of He_2^+ is 0.5

89. **Statement-I** : NaCl exist in hydrated form like $\text{NaCl} \cdot 10\text{H}_2\text{O}$.

Statement-II : LiCl exist in hydrated form like $\text{LiCl} \cdot 6\text{H}_2\text{O}$.

- (1) Statement-I is correct but statement-II is wrong
 (2) Both statement-I and II correct
 (3) Statement-I is wrong and statement-II is correct
 (4) Both statement-I and II incorrect.

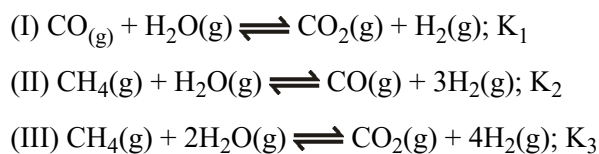
90. Incorrect order of bond energy.

- (1) $\text{Cl}_2 > \text{F}_2 > \text{Br}_2 > \text{I}_2$
 (2) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
 (3) $\text{N}_2 > \text{O}_2 > \text{H}_2 > \text{F}_2$
 (4) All of above

91. How many species does not exist is given :-
 SiF_6^{-2} , AlCl_6^{-3} , BrF_7 , BeF_4^{-2} , AlF_6^{-3} , PCl_6^{-}

- (1) 4 (2) 3
 (3) 1 (4) 2

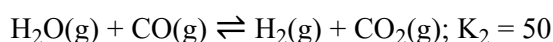
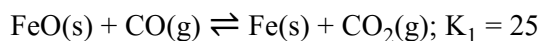
92. For the following three reactions I, II and III, equilibrium constants are given :-



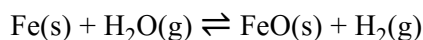
Which of the following relations is correct ?

- (1) $K_1 \sqrt{K_2} = K_3$ (2) $K_2 K_3 = K_1$
 (3) $K_3 = K_1 K_2$ (4) $K_3 = K_2^2 \cdot K_1$

93. Consider the following equilibrium



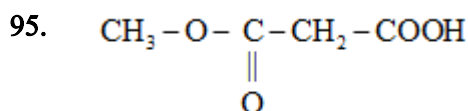
then equilibrium constant for reaction



- (1) $\frac{1}{2}$ (2) 2
 (3) $\sqrt{2}$ (4) 1

94. For which reaction $\log \frac{K_P}{K_C} - 2 \log RT = 0$

- (1) $\text{CO}_2\text{(g)} + \text{C(s)} \rightleftharpoons 2\text{CO(g)}$
 (2) $\text{CaCO}_3\text{(s)} \rightleftharpoons \text{CaO(s)} + \text{CO}_2\text{(g)}$
 (3) $\text{CO(g)} + \text{H}_2\text{O(g)} \rightleftharpoons \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$
 (4) $2\text{NH}_3\text{(g)} \rightleftharpoons \text{N}_2\text{(g)} + 3\text{H}_2\text{(g)}$



The correct systematic name of the above compound is :

- (1) 2-acetoxy ethanoic acid
 (2) 2-methoxy carbonyl ethanoic acid
 (3) 3-methoxy formyl ethanoic acid
 (4) 2-methoxy formyl acetic acid

96. Compound which can show G.I.?

- (1) 2-phenyl prop-1-ene
 (2) 1,1-dichloro prop-1-ene
 (3) 4-phenyl but-1-ene
 (4) 1-phenyl but-2-ene

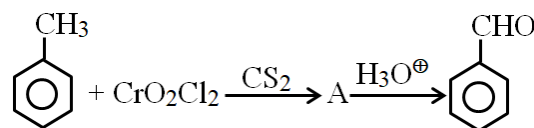
97. The most stable conformer of n-butane is :-

- (1) Gauche form (2) Partialy Eclipse form
 (3) Fully Eclipse form (4) Staggered form

98. Mesotartaric acid is optically inactive due to :-

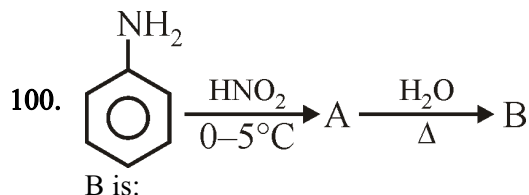
- (1) two chiral C-atoms
 (2) absence of element of symmetry
 (3) external compensation
 (4) internal compensation

99.



What is structure of chromium complex A ?

- (1)
 (2)
 (3)
 (4)



- B is:
 (1) Benzyl alcohol
 (2) Phenol
 (3) Benzene diazonium salt
 (4) Nitrobenzene

Topic : RESPIRATION SYSTEM, EXCRETORY SYSTEM, ENDOCRINE SYSTEM, MOLECULAR BASIS OF INHERITANCE (UP TO TRANSCRIPTION) [BACK UNIT] : HUMAN REPRODUCTION & REPRODUCTIVE HEALTH

SECTION-A (BIOLOGY-I)

101. How many total base pairs are present in one coil of DNA ?

- (1) 20 (2) 10 (3) 34 (4) 5

102. According to RNA world :

- (a) RNA molecule evolved from DNA
(b) RNA cannot act as catalyst
(c) DNA evolved from RNA
(d) RNA was first genetic material
How many statements are true ?

- (1) One (2) Two (3) Three (4) Four

103. In prokaryotes, which factor helps RNA polymerase enzyme to recognise the terminator site ?

- (1) ρ -factor (2) σ -factor
(3) α -factor (4) γ -factor

104. Match column I with column II and choose correct answer :-

Column I		Column II	
a	RNA polymerase I	i	hn-RNA
b	RNA polymerase II	ii	t-RNA
c	RNA polymerase III	iii	28sr-RNA

- (1) a-i, b-ii, c-iii (2) a-ii, b-i, c-iii
(3) a-iii, b-i, c-ii (4) a-iii, b-ii, c-i

105. **Assertion:** There is uniform distance between two strands of DNA.

Reason: Purine always comes opposite to and pairs with pyrimidine and vice-versa.

- (1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.

106. **Assertion :** RNA is better for the transmission of genetic information.

Reason : RNA can directly code for synthesis of protein.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

107. **Statement-I :** In RNA every nucleotide residue has an -OH group present at 2'-position in the ribose.

Statement-II : In RNA the uracil is found at the place of 5-methyl uracil.

- (1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II is incorrect.

108. Given below are two statements :

Statement-I :- The negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called nucleoid.

Statement-II :- The beads-on-string structure in chromatin is packaged to form chromatin fibers that are further coiled and condensed at metaphase stage of cell division to form chromosomes.

In the light of the above statements, choose the most appropriate answer from the options given below :-

- (1) Both statement-I and statement-II are incorrect
(2) Statement-I is correct but statement-II is incorrect
(3) Statement-I is incorrect but statement-II is correct
(4) Both statement-I and statement-II are correct

109. The segment of DNA coding for a polypeptide is known as :-
- (1) Operon (2) Recon
(3) Muton (4) Cistron
110. In DNA, promoter is the site for the initiation of :-
- (1) Replication
(2) Transcription
(3) Translation
(4) Both (1) & (2)
111. Which of the following steps in transcription is catalysed by RNA polymerase alone ?
- (1) Initiation (2) Elongation
(3) Termination (4) All of these
112. In Hershey & Chase experiment radioactive S^{35} detected in :-
- (1) Supernatant (2) Pellet
(3) Both (1) and (2) (4) Bacteria
113. In Avery, Macleod and Mc Carty experiment, which enzyme inhibits transformation?
- (1) DNase (2) RNase
(3) Protease (4) Gyrase
114. In a *E.coli* 4.6×10^6 bps are present then how many nucleosomes are present in it ?
- (1) 2.3×10^4 (2) 2.3×10^8
(3) 1.65×10^7 (4) Zero
115. When chromatin is seen under electron microscope the "beads-on-string" structure appear. These structures are :-
- (1) Spliceosome
(2) Kinetochore
(3) Nucleosome
(4) Nucleoside
116. Select the incorrect statement regarding DNA replication ?
- (1) Leading strand is formed in $5' \rightarrow 3'$ direction.
(2) The Okazaki fragments of lagging strand are formed in $3' \rightarrow 5'$ direction
(3) The leading strand is formed on the template strand with polarity $3' \rightarrow 5'$
(4) The lagging strand is formed on the template strand with polarity $5' \rightarrow 3'$
117. The promoter site and terminator site of transcription unit are located at :-
- (1) 3' end and 5' end of coding strand
(2) 3' end and 5' end of structural gene
(3) 5' (upstream) end and 3' (downstream) end of structural gene.
(4) 5' end (upstream) of structural gene only
118. The Okazaki fragments of discontinuous strand are joined by which of the following enzyme ?
- (1) DNA ligase (2) DNA gyrase
(3) Endonuclease (4) Primase
119. Which of the following is correct statement ?
- (1) RNA directly codes for protein
(2) Viruses having RNA genomes evolve slowly
(3) Proteins can generate their replicas through replication
(4) DNA is better than RNA for direct transmission of genetic information
120. In Meselson and Stahl's experiment ^{15}N was used as :-
- (1) Hybrid isotope
(2) Radioactive isotope
(3) Heavy isotope
(4) Light isotope

121. DNA replication of leading strand is-
- (1) Conservative (2) Continuous
(3) Semi-discontinuous (4) Discontinuous
122. 5' end of Okazaki fragments prior to removal of RNA primer consists of ?
- (1) Deoxyribonucleotides
(2) Ribonucleotides
(3) Both (1) and (2)
(4) Only AMP
123. Which of the following is common in both DNA and RNA ?
- (1) Hexose sugar (2) Histamine
(3) Thymine (4) Phosphate
124. In a double stranded DNA amount of Adenine is 35%, then what will be the amount of Guanine ?
- (1) 15 % (2) 30 % (3) 70 % (4) 35%
125. Which pyrimidine is common for both DNA and RNA ?
- (1) Adenine (2) Guanine
(3) Cytosine (4) Uracil
126. Choose the correct option :-
- (1) Larynx is a muscular box which helps in sound production
(2) Cartilaginous flap, epiglottis stop the entry of food into the larynx
(3) Trachea divides into 5th cervical vertebrae in right and left primary bronchi
(4) Trachea, Primary, Secondary, Tertiary and terminal bronchioles are supported by incomplete cartilaginous ring.
127. A process by which we intake O_2 rich air and expellout the CO_2 rich air is known as -
- (1) Respiration (2) Inspiration
(3) Breathing (4) Expiration

128. Select mismatch.

(1)	Asthma	Allergic disease characterised by inflammation of respiratory tract which leads to wheezing
(2)	Emphysema	Caused mainly due to cigarette smoking
(3)	Asbestosis	Leads to fibrosis of lungs
(4)	Pleurisy	Pericardial fluid decreases

129. **Statement-I** : Exhalation becomes active process only during forceful breathing.

Statement-II : Binding of oxygen with haemoglobin is called oxygenation.

- (1) Both statement-I and statement-II are correct.
(2) Both statement-I and statement-II are incorrect.
(3) Statement-I is correct but statement-II is incorrect.
(4) Statement-I is incorrect but statement-II is correct.
130. Oxygen binding to haemoglobin in blood is :
- (1) Directly proportional to the concentration of CO_2 in the medium.
(2) Inversely proportional to the concentration of CO_2 in the medium.
(3) Directly proportional to the concentration of CO in the medium.
(4) Independent of the concentration of CO in the medium.
131. Total percentage of O_2 transported by haemoglobin or RBC is :
- (1) 3% (2) 97%
(3) 49% (4) 25%

132. Read the following statements and choose the correct option.

Statement A : Thoracic chamber is an anatomically air tight chamber.

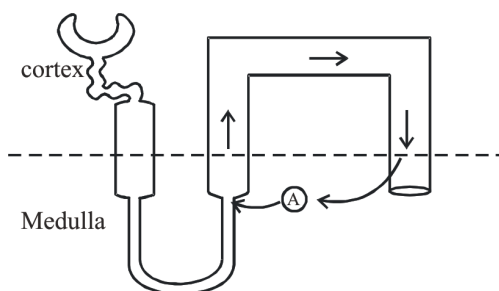
Statement B : We can alter pulmonary volume directly, without any change in the thoracic volume.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both statements A and B are correct
- (4) Both statements A and B are incorrect

133. Lungs are made up of air-filled sacs, the alveoli. They do not collapse even after forceful expiration because of :-

- (1) Residual volume
- (2) Inspiratory reserve volume
- (3) Tidal volume
- (4) Expiratory reserve volume

134.



In the given figure substance A name and its work is :-

- (1) NH_3 - Increases osmolarity in medullary part
- (2) NaCl - Increases osmolarity in medullary part
- (3) Urea - Increases osmolarity in medullary part
- (4) H_2O - Increases osmolarity in medullary part

135. In interstitial fluid of the kidney an increasing osmolarity from cortex to the inner medulla. This gradient is mainly caused by :

- (1) Na^+ , K^+
- (2) NaCl and Urea
- (3) NaCl , Water, HCO_3^-
- (4) Urea, K^+ , HCO_3^-

TG: @Chalnaayaaar
SECTION-B (BIOLOGY-I)

136. Base ratio of a DNA is 0.03 and the amount of A–T contents is 30000, than what is the amount of G–C contents in this DNA

- (1) 1000
- (2) 10000
- (3) 100000
- (4) 1000000

137. A normal DNA molecule is continuously replicated in N^{15} medium then what is the percentage of pure light DNA molecules in 4th generation.

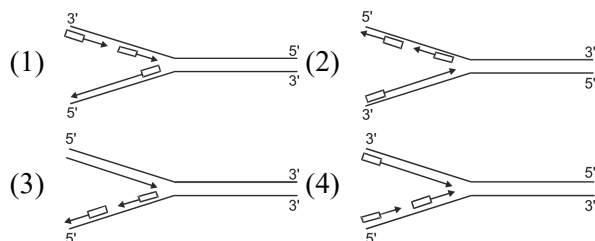
- (1) 12.5%
- (2) 25%
- (3) 0%
- (4) 6.25%

138. Choose the incorrect statement from the following statements (A – D) :-

- A) Deoxyribonucleoside triphosphates serves dual purpose.
- B) On template $5' \rightarrow 3'$, DNA replicates in continuous manner.
- C) DNA polymerase has to catalyse the reaction with low degree of accuracy.
- D) Any mistake during replication would result into mutations.

- (1) A and C
- (2) B and C
- (3) A and B
- (4) B and D

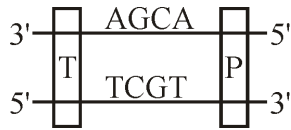
139. Which of the following diagram correctly represents the replication of DNA :-



140. During processing of hn-RNA in eukaryotes after transcription, the following event is true :-

- (1) In tailing, adenylate residues are added at 5' end in a template dependent manner
- (2) The process of splicing represents the dominance of RNA world
- (3) The split gene arrangements represents probably an advanced feature of genome
- (4) In prokaryotes RNA polymerase II transcribe hn-RNA

141. Find the sequence of mRNA encoded by following DNA sequence ?



- (1) 5' ACGA 3'
 (2) 5' AGCA 3'
 (3) 5' UGCU 3'
 (4) 5' UCGU 3'
142. A 100 bp linear ds DNA is shortened by 5 turns. What is the length of the remaining DNA ?
 (1) 34 nm
 (2) 17 nm
 (3) 68 nm
 (4) 136 nm
143. A small segment of DNA having a single ori has 5 Okazaki fragments each on either side of ori. Find the number of RNA primers used in the replication of DNA ?
 (1) 10 (2) 12 (3) 14 (4) 20
144. If $A + \frac{G}{C} + T = 51.5$ for one strand of DNA having 40 Adenine bases, 30 Guanine bases and 20 cytosine bases. Find $G + \frac{A}{T} + C$ for other strand of DNA ?
 (1) 60 (2) 62.5
 (3) 80.5 (4) 50.25
145. One of the ____ (i) ____ structures exhibited by DNA is the famous Watson and Crick model and the salient features of it are features of ____ (ii) ____ type DNA ? Fill in the blanks.
 (1) (i)-Primary, (ii)-B
 (2) (i)-Secondary, (ii)-A
 (3) (i) Secondary, (ii) -C
 (4) (i) Secondary, (ii) - B

146. Which is a common passage for food and air ?
 (1) Trachea
 (2) Oesophagus
 (3) Pharynx
 (4) Glottis
147. Arrange the following steps of respiration
 a. Utilisation of O_2 by cells
 b. Breathing
 c. diffusion of O_2 and CO_2 between blood and tissues
 d. transport of gases by the blood
 e. diffusion of gases across alveolar membrane
 (1) b e d a c
 (2) b d e a c
 (3) b e d c a
 (4) b d c a e
148. Thoracic cage in human beings is made up of following bones :
 (1) Ribs, vertebral column and diaphragm
 (2) Ribs, diaphragm and collar bone
 (3) Vertebral column, diaphragm and sternum
 (4) Ribs, vertebral column and sternum
149. The main function of loop of Henle is –
 (1) Passage of urine
 (2) Filtration of blood
 (3) Formation of urine
 (4) Conservation of water
150. Which of the following statement is incorrect?
 (1) Renin prevents conversion of angiotensinogen in blood to angiotensin
 (2) Aldosterone facilitates water reabsorption
 (3) ANF reduces sodium reabsorption
 (4) Renin increase-GFR

Topic : RESPIRATION SYSTEM, EXCRETORY SYSTEM, ENDOCRINE SYSTEM, MOLECULAR BASIS OF INHERITANCE (UP TO TRANSCRIPTION) [BACK UNIT] : HUMAN REPRODUCTION & REPRODUCTIVE HEALTH

SECTION-A (BIOLOGY-II)

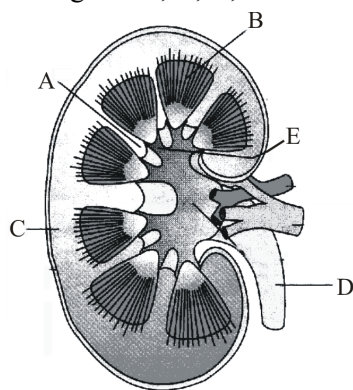
151. Which of the following causes an increase in sodium reabsorption in distal convoluted tubule?

- (1) Increase in aldosterone levels
- (2) Decrease in antidiuretic hormone levels
- (3) Decrease in aldosterone levels
- (4) Decrease in angiotensin levels

152. In micturition :-

- (1) Urethral sphincter contracts, urinary bladder relaxes
- (2) Urethral sphincter and urinary bladder contracts
- (3) Urethral sphincter relaxes, urinary bladder contracts
- (4) Urethral sphincter and urinary bladder relaxes

153. In the given diagram A, B, C, D & E are -



- (1) A - Renal column, B - Medullary pyramid, C - Cortex, D - Ureter, E - Renal calyx
- (2) A - Medullary pyramid, B - Renal column, C - Cortex, D - Ureter, E - Renal calyx
- (3) A-Renal column, B - Ureter, C - Cortex, D - Medullary pyramid, E - Renal pelvis
- (4) A - Cortex, B - Medullary pyramid, C - Renal column, D - Ureter, E - Hilum

154. Mark the incorrect statement :-

- (1) Average weight of each human kidney is 120-170 gm.
- (2) Antennal glands perform the excretory function in prawn
- (3) Cortex extends in between the medullary pyramids to form columns of bellini
- (4) Each nephron has two part glomerulus and the renal tubule

155. Which of the following Nitrogenous waste eliminated through sweat ?

- (1) Lactic acid
- (2) Urea
- (3) NaCl
- (4) All of the above

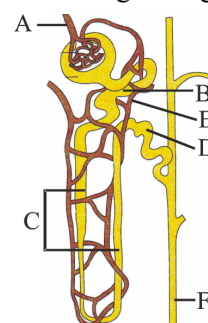
156. Which of the following are actively secreted into filtrate by tubular cells of nephron :-

- (1) Na^+ , K^+ , NH_3
- (2) K^+ , NH_3 , H^+
- (3) H^+ , KCO_3^+ , K^+
- (4) H^+ , NH_3 , Na^+

157. HCO_3^- is reabsorbed by which part of nephron :-

- (1) PCT, DCT
- (2) PCT, Henle's loop
- (3) DCT, Henle's loop
- (4) Collecting duct, Bowman's capsule

158. Select the option that correctly identifies the parts labelled from A to F in the given figure of nephron :-



- (1) A-Afferent arteriole, B-PCT, C-Henle's loop, D-DCT, E-Peritubular Capillary, F-Collecting duct
- (2) A-Efferent arteriole, B-PCT, C-Henle's loop, D-DCT, E-Collecting duct, F-Vasa recta
- (3) A-Afferent arteriole, B-Peritubular capillaries, C-Henle's loop, D-DCT, E-PCT, F-Collecting duct
- (4) A-Afferent arteriole, B-Henle's loop, C-Collecting duct, D-PCT, E-DCT, F-Peritubular capillaries

TG: @Chalnaayaaar

PHASE - MEC

1015CMD303361230059

14-08-2023

159. Life saving hormone of adrenal cortex is not related with :-

- (1) Increase gluconeogenesis
- (2) Increase lipogenesis
- (3) Increase lipolysis, increase erythropoiesis
- (4) Increase protein breakdown

160. The position of adrenal gland is

- (1) Anterior part of each kidney
- (2) Posterior part of each kidney
- (3) Ventral part of each kidney
- (4) Dorsal part of each kidney

161. A person is suffering from abnormal high blood pressure which hormone of body decreases blood pressure in following options:-

- (1) A steroid hormone secreted from adrenal medulla
- (2) A peptide hormone secreted from atria of heart
- (3) A protein hormone secreted from pineal gland
- (4) A fatty acid derivative hormone secreted from gonads

162. Which of the following does not secrete any hormone :-

- (1) Testes
- (2) Spleen
- (3) Stomach
- (4) Intestine

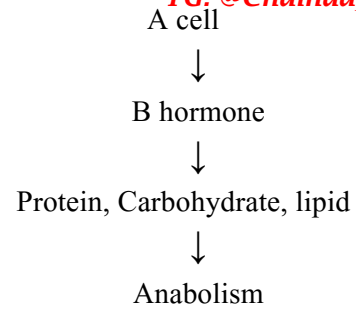
163. Read the following statements :-

- (A) Stimulates proteolysis
- (B) Provide relief when used in allergic disease
- (C) Can be used as immunosuppressive agent after kidney transplantation

Which of the following chemical performs all the above functions.

- (1) Thyroxine
- (2) Prostaglandin
- (3) Cortisol
- (4) Adrenaline

164.



Identify the A and B :-

- (1) A-Somatotroph cell, B → Growth hormone
- (2) A-Adrenal cortex cell, B → glucocorticoid
- (3) A-Beta cell, B → Insulin
- (4) A-Alpha cell, B → Glucagon

165. Intracellular receptors are found for

- a. Protein/peptide hormones
- b. Steroid hormones
- c. Iodothyronines

- (1) a and c
- (2) a and b
- (3) b and c
- (4) a only

166. Low secretion of growth hormone results in:

- (1) Gigantism
- (2) Acromegaly
- (3) Cretinism
- (4) Dwarfism

167. Basal metabolic rate is under control of :-

- (1) Glucagon
- (2) Insulin
- (3) Thyroxine
- (4) PTH

168. Arrange following events of female reproductive cycle in (human) in sequence :-

- (I) Secretion of FSH
- (II) Formation of corpus luteum
- (III) Growth of follicles
- (IV) Ovulation
- (V) Sudden increase in level of LH.

- (1) III → I → IV → II → V
- (2) I → III → V → IV → II
- (3) I → IV → III → V → II
- (4) II → I → III → IV → V

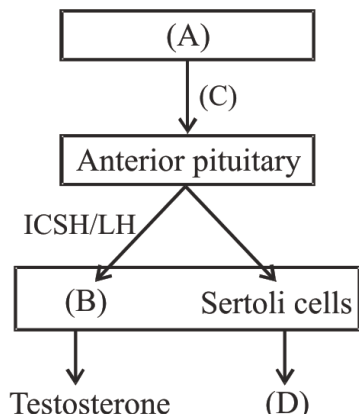
169. Each ovary is connected to the pelvic wall and the uterus by :-

- (1) Peritoneum
- (2) Ligaments
- (3) Adipose tissue
- (4) Areolar tissue

170. Extra embryonic membranes of mammalian embryo are derived from :-

- (1) Trophoblast (2) Inner cell mass
(3) Formative cells (4) Follicular cells

171.



In above chart identify A, B, C and D :-

	A	B	C	D
(1)	Germ cell	Leydig's cell	ABP	GnRH
(2)	Hypothalamus	Leydig's cells	GnRH	ABP
(3)	Hypothalamus	Germ cell	FSH	LH
(4)	Hypothalamus	Leydig's cell	FSH	ABP

172. Given below are four statements (a-d) regarding assisted reproductive technologies:-

- (a) **ZIFT** - The zygote or early embryo (with upto 8 blastomere) transferred into the fallopian tube
(b) **ICSI** - A sperm is directly injected into the fallopian tube to form an embryo in the laboratory
(c) **IUI** - The semen collected either from the husband or healthy donor is artificially introduced into the uterus
(d) **GIFT** - Transfer of zygote collected from a donor into the fallopian tube of another female who cannot produce one but can provide suitable environment for fertilisation and development
Which two of the above statements are correct?

- (1) a and d (2) b & c
(3) a & c (4) b & d

173. Choose the incorrect match pair :-

(1)	Multiload 375	Suppress sperm motility
(2)	Progestasert	Make uterus unsuitable for implantation.
(3)	Lippes loop	Make cervix hostile to the sperm
(4)	CuT	Suppress sperm motility

174. Read the following statements carefully and choose the correct one :-

- (1) Saheli is a steroidal pill.
(2) Multiload-375 is a hormone releasing IUD
(3) i-pill used as a emergency contraceptive pill.
(4) Barrier methods prevent the ovulation.

175. are also essential in certain cases where continuation of pregnancy could be harmful or even fatal either to the mother or to the foetus or both:-

- (1) Medical termination of Pregnancy
(2) Intra Uterine devices
(3) Contraceptive pills
(4) Female condom

176. Which of the following method of contraception is effective only upto a maximum period of six months following parturition :-

- (1) Cu T
(2) LNG-20
(3) Coitus interrupts
(4) Lactational amenorrhoea

177. Which of the following contraceptive method has its additional benefit of protecting the user from contracting STD's and AIDS ?

- (1) IUD (2) Condom
(3) Hormonal Oral pill (4) Implant

178. Effect of the hormone releasing IUDs compared to copper releasing IUDs is :-

- (1) Suppressing the sperm motility
- (2) Reducing the fertilising capacity of sperm
- (3) Making the uterus unsuitable for implantation and cervix hostile to sperm
- (4) Increasing phagocytosis of sperm within the uterus

179. In humans, at the end of the first meiotic division, male germ cells form :-

- (1) Spermatogonia
- (2) Primary spermatocytes
- (3) Secondary spermatocytes
- (4) Spermatids

180. Identify the hormones that are secreted in large amount prior to ovulation.

- (A) LH (B) FSH
(C) Progesterone (D) Estrogen

- (1) Only A (2) Only A & D
- (3) A, B & D (4) A, B, C

181. Given below some Assisted Reproductive Technologies (ART). In which find out invitro fertilization (IVF) :-

- (a) ZIFT (b) ICSI (c) AI
(d) IUT (e) GIFT

- (1) a, b, c, e (2) a, c, e
- (3) a, b, d (4) a, b, c, d, e

182. Select wrong statements :-

- (1) One ovum is released during the middle of each menstrual cycle.
- (2) Menstruation occur only if the released ovum is not fertilized.
- (3) Lack of menstruation always will be indicator of pregnancy.
- (4) The secretion of gonadotropins increase gradually during the follicular phase.

183. **Statement-I** : Parturition is induced by a complex neuroendocrine mechanism.

Statement-II : Placenta is jointly formed by chorionic villi and uterine tissue.

- (1) Both statement-I and statement-II are correct
- (2) Statement-I is correct but statement-II is incorrect
- (3) Statement-I is incorrect but statement-II is correct
- (4) Both statement-I and statement-II are incorrect

184. **Assertion (A)** : Degeneration of endometrium leads to menstruation.

Reason (R) : In the absence of fertilisation, the corpus luteum is maintained by HCG hormone.

- (1) Both (A) & (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) & (R) are correct and (R) is the correct explanation of (A)

185. Which of the following statement is true ?

- (1) Acrosome of sperm serves no particular function.
- (2) Menstrual fluid can easily clot.
- (3) The part of the fallopian tube closest to the ovary is ampulla.
- (4) Viability of sperm is determined by it's motility.

SECTION-B (BIOLOGY-II)

186. Incorrect match is:

- (1) Glomerulonephritis : Inflammation of glomeruli of kidney
- (2) Renal calculi : Stones of oxalates within kidney
- (3) Ketonuria : Ketone bodies in urine
- (4) Glycosuria : High urea in blood

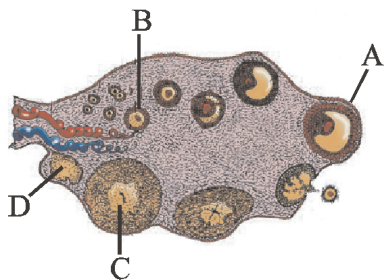
187. Milk formation is regulated by :

- (1) Oxytocin
- (2) Follicle stimulating hormone
- (3) Luteinizing hormone
- (4) Prolactin

188. Maintenance of H_2O and electrolytes balance is also influenced by :-

- (1) Cortisol
- (2) Insulin
- (3) Thyroxine
- (4) Adrenaline

189. The figure below shows development of follicles (A, B, C, D). Select the option which responsible for maintenance of pregnancy:-



- (1) B
- (2) D
- (3) A
- (4) C

190. Placenta is made up of the tissue :-

- (1) Only maternal tissue
- (2) Only embryonic tissue
- (3) Both maternal and embryonic tissue
- (4) Only by embryoblast

191. The reproductive cycle of ____ (A) ____ is called menstrual cycle

Fill the blanks with suitable option.

- (1) Female primates
- (2) Female non primates
- (3) All primates
- (4) All non primates

192. The correct sequence of spermatogenesis stage leading to the formation of sperms in a mature human testis is :-

- (1) Spermatogonia-spermatocytes-spermatids-sperms
- (2) Spermatids-spermatocytes-Spermatogonia-sperms
- (3) Spermatogonia-spermatids-spermatocytes-sperms
- (4) Spermatocytes-Spermatogonia-spermatids-sperms

193. Which of the following contraceptive method is least effective ?

- (1) Withdrawal
- (2) IUDS
- (3) Oral contraceptives
- (4) Condoms

194. RCH stands for

- (1) Reproductive and Clinical Health
- (2) Regular Contraceptive Hormonal method
- (3) Reproductive and Child Health care programme
- (4) Research Centre for Hormonal contraceptives

195. The signals for parturition originate from the :-

- (A) Fully developed foetus
- (B) Placenta
- (C) Uterus
- (D) Ovary

- (1) A & B
- (2) A & D
- (3) C & D
- (4) B & D

196. The main parturition hormone is:

- (1) Oxytocin
- (2) Progesterone
- (3) Estrogen
- (4) LH

197. Function of placenta is :-

- (1) Supply of O_2 to embryo
- (2) Produces several hormones during pregnancy.
- (3) Removal of CO_2 produced by the embryo
- (4) All of above

198. Which of the following is not a correct statement about umbilical cord ?

- (1) It connects the placenta to the embryo.
- (2) It has 100% foetal blood.
- (3) It helps in the transport of substances to and from the embryo.
- (4) It produce several hormones like hPL, estrogen and progesterone.

199. Which of the following structure is essential for maintenance of the endometrium ?

- (1) Graafian follicle
- (2) Corpus albicans
- (3) Corpus luteum
- (4) Granulosa cells

200. What is the effect of menopause on the levels of FSH, LH, estrogen and GnRH :-

- (1) [FSH]↑; [LH]↑; [Estrogen]↓; [GnRH]↑
- (2) [FSH]↓; [LH]↓; [Estrogen]↓; [GnRH]↓
- (3) [FSH]↓; [LH]↓; [Estrogen]↑; [GnRH]↑
- (4) [FSH]↓; [LH]↑; [Estrogen]↓; [GnRH]↓

 QUESTIONS BREAK-UP CHART FOR MINOR TEST FROM CURRENT & PREVIOUS UNITS (Session : 2023 - 2024)			
ENTHUSE ADV. COURSE : MEC			
MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
07 10-09-2023	SYLLABUS COVERED FROM 16-08-2023 TO 09-09-2023	P	CAPACITOR & WORK, ENERGY AND POWER
		C	GOC - I
		B	LIFE CYCLE OF ANGIOSPERM & REPRODUCTION IN ORGANISM

TG: @Chalnaayaaar

TG: @Chalnaayaaar

ALLEN[®] CAREER INSTITUTE Pvt. Ltd.

Registered & Corporate Office : 'SANKALP', CP-6, Indra Vihar, Kota (Rajasthan) INDIA-324005
Ph. : +91-744-3556677, +91-744-2757575 | E-mail : info@allen.ac.in | Website : www.allen.ac.in